

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
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Course Code	24CS002PC215	Course Title	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment	Week11 - Tuesday	Time(s)	
Duration	2 Hours	Applicable to Batches	
AssignmentNumber:22.1(Present assignment number)/24(Total number of assignments)			
1	Question		Expected Time to complete
	Lab 22 – Advanced Ethical Considerations: Addressing Complex Scenarios and Limitations Lab Objectives:		Week 11 - Monday

• Understand ethical concerns in AI-assisted coding. • Explore complex real-world scenarios where AI decisions have consequences. • Learn to identify limitations and biases in AI-generated solutions. • Encourage critical thinking and human responsibility in AI collaboration.	
Task 1 – AI and Inclusivity • Ask AI to generate a chatbot for customer support. • Test if it handles multiple languages or accessibility features (e.g., screen readers). • Discuss: How lack of inclusivity creates digital inequality. Ethical ways to ensure accessibility in AI-generated systems PROMPT : Generate a chatbot for customer support and also test if it handles multiple languages or accessibility features (e.g., screen readers). Also discuss how lack of inclusivity creates digital inequality. CODE:	

```

def simple_chatbot(user_input, conversation_history=None):
    """
    A basic customer support chatbot function.

    Args:
        user_input: The user's query (string).
        conversation_history: A list of previous user inputs and chatbot responses (optional).

    Returns:
        A string containing the chatbot's response.
    """
    if conversation_history is None:
        conversation_history = []

    # Simple keyword matching for responses
    if "hello" in user_input.lower() or "hi" in user_input.lower():
        response = "Hello! How can I help you today?"
    elif "bye" in user_input.lower() or "goodbye" in user_input.lower():
        response = "Goodbye! Have a great day."
    elif "support" in user_input.lower() or "help" in user_input.lower():
        response = "I can assist you with common issues. Please tell me more about your problem."
    elif "return" in user_input.lower() or "refund" in user_input.lower():
        response = "For returns and refunds, please visit our returns policy page on our website."
    elif "order status" in user_input.lower():
        response = "To check your order status, please provide your order number."
    else:
        response = "I'm sorry, I don't understand. Could you please rephrase your question?"

    # Simple mechanism to remember conversation history (optional)
    conversation_history.append({"user": user_input, "chatbot": response})

    return response, conversation_history

# Example usage (for testing the function)
# history = []
# print(simple_chatbot("Hello", history))
# print(simple_chatbot("I need support", history))
# print(simple_chatbot("What about a refund?", history))
# print(simple_chatbot("Tell me about my order status", history))
# print(simple_chatbot("Some random query", history))

```

```

test_phrases = [
    "Hello", # English
    "Hola", # Spanish
    "Bonjour", # French
    "Hallo", # German
    "Support", # English
    "Apoyo", # Spanish
    "Soutien", # French
    "Unterstützung", # German
    "Refund", # English
    "Reembolso", # Spanish
    "Remboursement", # French
    "Rückerstattung" # German
]

for phrase in test_phrases:
    response, _ = simple_chatbot(phrase)
    print(f"Input: {phrase}")
    print(f"Chatbot Response: {response}")
    print("-" * 20)

```

Input: Hello
Chatbot Response: Hello! How can I help you today?

Input: Hola
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Bonjour
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Hallo
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Support
Chatbot Response: I can assist you with common issues. Please tell me more about your problem.

Input: Apoyo
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Soutien
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Unterstützung
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Refund
Chatbot Response: For returns and refunds, please visit our returns policy page on our website.

Input: Reembolso
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Remboursement
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

Input: Rückerstattung
Chatbot Response: I'm sorry, I don't understand. Could you please rephrase your question?

```
6 # Simulate screen reader interaction and analyze accessibility
def simulate_screen_reader(chatbot_response):
    """
    Simulates screen reader output for a given chatbot response.
    In a real scenario, screen readers read out text content.
    This function simply returns the response as it would be read.
    """
    return chatbot_response

# Example interactions
user_inputs = [
    "Hello",
    "I need help",
    "How do I return an item?",
    "What's my order status?",
    "Tell me something random."
]

print("--- Screen Reader Simulation and Accessibility Analysis ---")

for user_input in user_inputs:
    chatbot_response = simple_chatbot(user_input)
    screen_reader_output = simulate_screen_reader(chatbot_response)

    print(f"User Input: {user_input}")
    print(f"Chatbot Response: {chatbot_response}")
    print(f"Simulated Screen Reader Output: {screen_reader_output}")

    # Basic analysis of potential accessibility issues
    print("Potential Accessibility Issues:")
    if "I'm sorry, I don't understand" in chatbot_response:
        print("- Ambiguous phrasing for screen reader users: The generic error message doesn't provide context on what the chatbot 'can' do.")
    # Add more analysis based on other response types if needed
    elif "Hello! How can I help you today?" in chatbot_response:
        print("- Clear and direct greeting.")
    elif "I can assist you with common issues. Please tell me more about your problem." in chatbot_response:
        print("- Provides context and prompts for more information.")
    elif "For returns and refunds, please visit our returns policy page on our website." in chatbot_response:
        print("- The returns and refunds, please visit our returns policy page on our website." in chatbot_response:
        print("- Before to an external resource. A screen reader user would need this link to be accessible (e.g., clearly labeled).")
    elif "To check your order status, please provide your order number." in chatbot_response:
        print("- Clear instruction on what information is needed.")

print("--- Discussion on Screen Reader Interpretation and Improvements ---")

print("How a screen reader would likely interpret the chatbot's responses:")
print("A screen reader would read the text of the chatbot's responses aloud. For the current simple_chatbot, the output is purely text-based. There are no interactive elements (like buttons or links) within the response itself that the screen reader would need to interpret.")
print("The interpretation would be straightforward: the text content of the response as a linear stream of words.")

print("Potential accessibility improvements:")
1. Provide more specific error messages: Instead of a generic "I don't understand," the chatbot could offer suggestions based on keywords it 'does' recognize or provide a list of commands it can handle.
2. Ensure linked content is accessible: If the chatbot directs users to a webpage (like the returns policy page), ensure that page is fully accessible with proper ARIA attributes, focus management, and semantic HTML.
3. Consider conversational flow while not directly a screen reader issue, a clear and predictable conversational flow benefits all users, including those using screen readers.
4. Use clear and concise language: Avoid jargon or overly complex sentences that might be difficult for a screen reader user to process.
5. If interactive elements are added later: Ensure buttons, links, or other interactive components within the chatbot interface have appropriate ARIA labels and roles to clearly convey their purpose to a screen reader.
```

discuss how lack of inclusivity creates digital inequality:

Digital Inequality and the Inclusivity Gap in Chatbots Digital inequality refers to the uneven distribution of access to, use of, and skills related to digital technologies. It's not just about whether someone has a computer or internet connection, but also about their ability to effectively utilize digital tools for education, employment, healthcare, and civic participation. A significant factor exacerbating digital inequality is the lack of inclusivity in the design and implementation of digital tools, such as customer support chatbots. As demonstrated by the previous testing, our simple chatbot lacked multilingual capabilities and had limitations in how its responses might be interpreted by assistive technologies. These limitations are representative of broader issues in the digital landscape. The lack of inclusivity in chatbots contributes to digital inequality in several ways: • Language Barriers: Chatbots that only support a limited number of languages exclude a significant portion of the global population. Users who are not proficient in the supported languages are unable to access information or receive support, effectively creating a digital divide based on linguistic background. This can prevent access to essential services, emergency information, or customer support for products they have purchased. • Accessibility for Users with Disabilities: As highlighted in the accessibility testing, the way a chatbot's responses are structured and presented can impact users who rely on screen readers or other assistive technologies. Ambiguous phrasing, lack of clear navigation to external resources, or the absence of alternative input methods (like voice commands) can make chatbots unusable for individuals with visual, auditory, cognitive, or motor impairments. This restricts their ability to independently interact with digital services. • Potential Biases in Responses: Chatbots trained on biased data can perpetuate and amplify societal inequalities. This can manifest in discriminatory language, unfair recommendations, or a lack of understanding of diverse cultural contexts. Such biases can lead to marginalized groups receiving inadequate or inappropriate support, further entrenching existing inequalities. These limitations can have tangible negative impacts on individuals and communities. For example, a non-English speaker might be unable to resolve a critical issue with a utility provider's chatbot, leading to service disruption. A user who is blind might be unable to use a healthcare provider's chatbot to schedule an appointment, hindering their access to medical care. To promote equitable access and participation in the digital world, it is crucial to design inclusive chatbots and other digital tools. This involves: • Prioritizing Multilingual Support: Developing chatbots that can understand and respond in multiple languages is essential for reaching diverse user populations. • Implementing Accessibility Standards: Designing interfaces and responses with accessibility guidelines (like WCAG) in mind ensures that users with disabilities can effectively interact with the chatbot. This includes providing clear and concise language, ensuring compatibility with screen readers, and offering alternative interaction methods where possible. • Addressing Algorithmic Bias: Rigorous testing and mitigation strategies are needed to identify and reduce biases in the data used to train chatbots, ensuring fair and equitable interactions for all users. • User Involvement in Design: Including individuals from diverse linguistic backgrounds, with varying abilities, and from different cultural contexts in the design and testing phases can help identify and address potential inclusivity gaps early on. By consciously designing for inclusivity, we can ensure that digital tools like chatbots become bridges rather than barriers, fostering a more equitable and accessible digital future for everyone.	
OBSERVATION : Based on the testing, the current chatbot lacks multilingual support and has basic accessibility. This highlights how the absence of these features contributes to digital inequality by creating barriers for users with different language needs and those relying on assistive technologies. Task 2 – Intellectual Property and Plagiarism • Ask AI to write a code snippet for image compression. • Check if AI reuses code from open-source libraries without attribution. Discuss: • How copyright violations affect developers. • Best practices for attribution when using AI-generated code. PROMPT : write a code snippet for image compression and Check if code is refused from open-source libraries without attribution. CODE:	

```

from PIL import Image
import os

# Replace with the path to your image file
input_image_path = 'input_image.jpg'
output_image_path = 'compressed_image.jpg'

# Create a dummy image file for demonstration if it doesn't exist
if not os.path.exists(input_image_path):
    # Create a simple red image
    dummy_image = Image.new('RGB', (100, 100), color = 'red')
    dummy_image.save(input_image_path)
    print(f"Created a dummy image at {input_image_path} for demonstration.")

try:
    # Open the image
    img = Image.open(input_image_path)

    # Get original file size
    original_size = os.path.getsize(input_image_path)

    # Save the image with compression (e.g., quality=50)
    img.save(output_image_path, quality=50)

    # Get compressed file size
    compressed_size = os.path.getsize(output_image_path)

    print(f"Original image size: {original_size} bytes")
    print(f"Compressed image size: {compressed_size} bytes")

except FileNotFoundError:
    print(f"Error: The file '{input_image_path}' was not found.")
except Exception as e:
    print(f"An error occurred: {e}")

```

```

... Created a dummy image at input_image.jpg for demonstration.
Original image size: 825 bytes
Compressed image size: 824 bytes

```

IMAGE COMPRESSION:

The Importance of Attribution When Using Open-Source Code

Open-source software is built on a foundation of collaboration and sharing. Developers contribute their time and expertise to create tools and libraries that are freely available for others to use, modify, and distribute. While this open model fosters innovation, it comes with responsibilities, one of the most significant being attribution.

Attribution means giving credit to the original authors or contributors of the open-source code you are using. It's not just a matter of good etiquette; it's often a legal requirement and a fundamental principle of the open-source community.

Why is Attribution Necessary?

- 1. Legal Requirements:** Most open-source licenses (such as MIT, Apache, GPL) explicitly require users to provide attribution. Failing to do so can be a breach of the license agreement, potentially leading to legal issues, including lawsuits and injunctions. These licenses are designed to protect the rights of the original creators while promoting the free use and distribution of their work.
- 2. Ethical Responsibility:** Even when a license doesn't strictly require it, providing attribution is an ethical obligation. It acknowledges the effort and skill of the developers who created the code you are benefiting from. Using someone else's work without credit is a form of intellectual dishonesty and undermines the spirit of collaboration that drives the open-source movement.
- 3. Supporting the Open-Source Community:** Attribution helps to promote the open-source projects and their contributors. It makes it easier for others to discover the original code, contribute to the project, or seek help from the community. It fosters a healthy ecosystem where contributions are recognized and valued.
- 4. Transparency and Reproducibility:** Proper attribution makes your own work more transparent and reproducible. It allows others to understand which components you used and where they came from, which is crucial for debugging, verification, and building upon your work.

Consequences of Failing to Provide Proper Attribution:

- **Legal Repercussions:** As mentioned, you could face legal action from the copyright holders for violating the terms of the open-source license. This could result in significant financial penalties and legal costs.
- **Damage to Reputation:** Failing to attribute code can severely damage your personal or organizational reputation within the developer community. It can lead to a loss of trust and make it difficult to collaborate on future projects.
- **Hindering the Collaborative Spirit:** When attribution is neglected, it discourages developers from contributing to open-source projects, as their work may be used without recognition. This can slow down innovation and the development of valuable tools.
- **Project Instability:** If your project relies on open-source code without proper attribution, it might be harder to maintain or update dependencies if you can't easily track the origin of the code.

Examples of Proper Attribution:

The specific requirements for attribution vary depending on the open-source license. However, common practices include:

- **Including the License File:** Distribute a copy of the open-source license with your project.
- **Adding a NOTICE File:** Create a NOTICE file that lists the open-source components used and their respective licenses and attribution requirements.
- **In-Code Comments:** Include comments in your code where you use snippets or functions from open-source libraries, referencing the source and license.
- **Documentation:** Clearly list the open-source dependencies and their attributions in your project's documentation.
- **About or Credits Section:** In user-facing applications, include an "About" or "Credits" section that lists the open-source software used.

For the image compression code using the Pillow library (which is released under the permissive HPK license), a simple attribution in your project's documentation or a NOTICE file would be appropriate, stating that your project uses Pillow and including a link to the Pillow documentation or repository.

By consistently practicing proper attribution, you contribute to a thriving and ethical open-source ecosystem while protecting yourself from potential legal and reputational harm.

Discuss How copyright violations affect developers :

How Copyright Violations Affect Developers

Copyright is a legal right that grants the creator of original works of authorship exclusive rights to use and distribute their work. In the context of software development, copyright protects the source code, the look and feel of the software, and other creative elements. Copyright violations, also known as infringement, occur when someone uses, reproduces, distributes, or creates derivative works from copyrighted material without the permission of the copyright holder.

Copyright violations can have significant negative impacts on developers:

- **Loss of Income:** For developers who create commercial software or sell their code, copyright infringement can lead to substantial financial losses. Unauthorized distribution or use of their work deprives them of potential revenue.
- **Reduced Motivation and Innovation:** When developers see their work being used without permission or compensation, it can be demotivating. This can stifle creativity and reduce the incentive to develop new and innovative software.
- **Difficulty in Maintaining and Improving Software:** If copyrighted code is copied and modified without proper authorization, it can lead to fragmented versions of the software. This makes it difficult for the original developer to provide support, updates, and improvements to their official version.
- **Legal Costs and Stress:** Pursuing legal action against copyright infringers can be a lengthy, expensive, and stressful process for developers. Even if they win the case, the time and resources spent can be considerable.
- **Damage to Reputation:** While less common for the copyright holder, if their work is associated with infringing or low-quality derivative works, it could indirectly impact their reputation.
- **Undermining the Open-Source Ecosystem:** While open-source licenses encourage sharing, they still have terms that must be followed. Copyright violations within the open-source community, such as failing to provide attribution, undermine the principles of collaboration and trust that are essential for its functioning.

In essence, copyright protection is crucial for developers to protect their intellectual property, ensure fair compensation for their work, and maintain control over how their software is used and distributed. Violations of copyright not only harm individual developers but also have broader negative consequences for the software development industry as a whole by discouraging innovation and undermining legal frameworks.

Discuss Best practices for attribution when using AI-generated code:

- The initial chatbot lacked multilingual support and had basic accessibility, contributing to digital inequality.
- Image compression code using Pillow was provided.
- Attribution for open-source code is legally and ethically required, and failure to do so can have negative consequences.
- Best practices for attributing AI-generated code include transparency, review, and understanding the tool's terms.
- Copyright violations harm developers through loss of income, reduced innovation, legal issues, and damage to reputation.

Task 3 – Dual-Use Code Risks

Use AI to generate a network scanning tool.

Discuss how the same tool can be used for ethical cybersecurity testing or illegal hacking.
Debate whether AI should refuse such requests, and where to draw the line.

PROMPT :

Generate a network scanning tool by python code and also discuss how the same tool can be used for ethical cybersecurity testing or illegal hacking and Debate whether AI should refuse such requests, and where to draw the line.

CODE:

```
from PIL import Image
import os

# Replace with the path to your image file
input_image_path = 'input_image.jpg'
output_image_path = 'compressed_image.jpg'

# Create a dummy image file for demonstration if it doesn't exist
if not os.path.exists(input_image_path):
    # Create a simple red image
    dummy_image = Image.new('RGB', (100, 100), color = 'red')
    dummy_image.save(input_image_path)
    print(f"Created a dummy image at {input_image_path} for demonstration.")

try:
    # Open the image
    img = Image.open(input_image_path)

    # Get original file size
    original_size = os.path.getsize(input_image_path)

    # Save the image with compression (e.g., quality=50)
    img.save(output_image_path, quality=50)

    # Get compressed file size
    compressed_size = os.path.getsize(output_image_path)

    print(f"Original image size: {original_size} bytes")
    print(f"Compressed image size: {compressed_size} bytes")

except FileNotFoundError:
    print(f"Error: The file '{input_image_path}' was not found.")
except Exception as e:
    print(f"An error occurred: {e}")
```

Created a dummy image at input_image.jpg for demonstration.
Original image size: 825 bytes
Compressed image size: 824 bytes

Ethical vs. Illegal Use of Network Scanning Tools

Network scanning tools are powerful utilities that can be used to probe a network and identify active hosts, open ports, and potentially vulnerable services. The legality and ethical implications of using these tools depend entirely on the intent of the user and whether they have proper authorization.

Ethical Cybersecurity Testing:

In the realm of ethical cybersecurity, network scanning is a fundamental and essential practice. Security professionals, often referred to as "ethical hackers" or "penetration testers," use these tools to:

- **Vulnerability Assessments:** Scan their own networks or networks they are explicitly authorized to test to identify potential weaknesses and security flaws before malicious actors do. This includes discovering open ports that shouldn't be exposed, identifying running services with known vulnerabilities, and mapping the network topology.
- **Penetration Testing:** Simulate real-world attacks to evaluate the effectiveness of security controls. Network scanning is often the initial reconnaissance phase of a penetration test, providing crucial information about the target environment.
- **Security Audits:** Verify that security policies and configurations are correctly implemented by scanning the network for compliance issues.
- **Network Inventory and Mapping:** Understand the devices connected to a network and how they are interconnected, which is vital for effective network management and security.

When used ethically, network scanning is performed with explicit permission from the network owner, is conducted within a defined scope, and the findings are used to improve security posture.

Illegal Hacking and Unauthorized Access:

Conversely, the same network scanning tools can be misused for malicious and illegal purposes. Unauthorized network scanning is often the first step taken by cybercriminals and malicious actors to:

- **Identify Vulnerable Targets:** Scan vast ranges of IP addresses or specific organizations to find systems with open ports, weak security configurations, or known vulnerabilities that can be exploited for unauthorized access.
- **Reconnaissance for Attacks:** Gather information about a target network's infrastructure, operating systems, and running services to plan more sophisticated attacks, such as exploiting specific software vulnerabilities or launching denial-of-service attacks.
- **Map Networks for Intrusion:** Understand the network layout to navigate and move laterally within a compromised network.
- **Discover Exposed Data:** Identify services or devices that are inadvertently exposing sensitive information.

Unauthorized scanning of networks is illegal in most jurisdictions and is considered a preparatory act for cybercrime. It violates privacy and security by probing systems without permission. The intent is typically to find entry points for unauthorized access, data theft, system disruption, or other malicious activities.

In summary, while the technical functionality of network scanning tools is neutral, their use is strictly divided along ethical and legal lines based on authorization and intent. Ethical use involves permission-based scanning for security improvement, while illegal use involves unauthorized scanning for malicious purposes.

OBSERVATION :

The network scanning tool works for basic checks, but its use highlights the dual nature of such tools – useful for ethical security testing but also applicable for illegal hacking. This raises a complex debate about where AI should draw the line on providing potentially harmful information.

Task 4 – Human Oversight in Healthcare

- Scenario: AI writes code for detecting heart anomalies from ECG data.
- Discuss:
 - Should doctors rely fully on AI outputs?
 - What oversight is mandatory before deployment in real hospitals?
 - How should responsibility be shared?

PROMPT :

Write a python code for detecting heart anomalies from ECG data.

CODE :

```

import pandas as pd
import numpy as np
from scipy.signal import butter, filtfilt

# Assume the ECG data is in a CSV file named 'ecg_data.csv'
# and the ECG signal is in a column named 'signal'.
try:
    df = pd.read_csv('ecg_data.csv')
except FileNotFoundError:
    # Create a dummy dataframe if the file is not found
    print("ecg_data.csv not found. Creating a dummy DataFrame.")
    data = {'signal': np.sin(np.linspace(0, 100, 1000)) + np.random.normal(0, 0.1, 1000)}
    df = pd.DataFrame(data)

# Bandpass filter
def butter_bandpass(lowcut, highcut, fs, order=5):
    nyq = 0.5 * fs
    low = lowcut / nyq
    high = highcut / nyq
    b, a = butter(order, [low, high], btype='band')
    return b, a

def bandpass_filter(data, lowcut, highcut, fs, order=5):
    b, a = butter_bandpass(lowcut, highcut, fs, order=order)
    y = filtfilt(b, a, data)
    return y

# Assuming a sampling frequency of 200 Hz
fs = 200.0
lowcut = 0.5
highcut = 40.0

if 'signal' in df.columns:
    df['filtered_signal'] = bandpass_filter(df['signal'], lowcut, highcut, fs)

    # Moving average noise reduction
    window_size = 10
    df['denoised_signal'] = df['filtered_signal'].rolling(window=window_size, center=True).mean()

    # Normalize the data using Min-Max scaling
    min_val = df['denoised_signal'].min()
    max_val = df['denoised_signal'].max()
    df['normalized_signal'] = (df['denoised_signal'] - min_val) / (max_val - min_val)

    display(df.head())
    display(df.tail())
else:
    print("Column 'signal' not found in the DataFrame.")

```

*** ecg_data.csv not found. Creating a dummy DataFrame.

	signal	filtered_signal	denoised_signal	normalized_signal
0	0.100277	-0.014750	NaN	NaN
1	0.062543	-0.004577	NaN	NaN
2	0.209723	0.022935	NaN	NaN
3	0.128826	0.090059	NaN	NaN
4	0.368380	0.211642	NaN	NaN
995	-0.642552	-0.600300	-0.579841	0.247847
996	-0.848832	-0.496667	NaN	NaN
997	-0.524402	-0.376029	NaN	NaN
998	-0.568949	-0.267583	NaN	NaN
999	-0.462557	-0.176686	NaN	NaN

Should doctors rely fully on AI outputs?

ANS: While AI can be a powerful tool to assist doctors in tasks like identifying patterns and improving efficiency, **doctors should not rely fully on AI outputs**. AI lacks the comprehensive understanding, clinical judgment, and consideration of individual patient context that a doctor provides. AI is best used as a tool to augment a doctor's expertise, not replace it.

What oversight is mandatory before deployment in real hospitals?
ANS : Before deploying AI in hospitals, mandatory oversight typically involves regulatory approval (like from the FDA in the US or similar bodies elsewhere), rigorous clinical validation and testing in real-world settings, ethical review, data privacy and security assessments (like HIPAA compliance), and clear protocols for human oversight and decision-making.

How should responsibility be shared?
ANS : Responsibility should be shared, with AI as a tool providing information, and the **doctor retaining the final decision-making authority and accountability** for patient care. Developers and institutions also share responsibility for the AI's safety, effectiveness, and ethical deployment.

Task 5 – AI in Decision-Making

- Ask AI to generate a loan approval system.
- Analyze if the decision process is explainable to users.
- Discuss:
 - What risks arise if AI decisions are a “black box”?
 - How explainability ensures fairness in financial decisions

PROMPT :

Generate a loan approval system and analyze if the decision process is explainable to users.

CODE:

```
import pandas as pd

# Load the dataset
df = pd.read_csv('https://raw.githubusercontent.com/drsrscientist/DSdata/master/loan_prediction.csv')

# Display the first few rows
display(df.head())
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

```
/tmp/ipython-input-1179260901.py:2: FutureWarning: inplace is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(col: value, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.
```

```
df[col].fillna(df[col].mode()[0], inplace=True)
/tmp/ipython-input-1179260901.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(col: value, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.
```

	0
Loan_ID	0
Gender	0
Married	0
Dependents	0
Education	0
Self_Employed	0
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	0
Loan_Amount_Term	0
Credit_History	0
Property_Area	0
Loan_Status	0

```
from sklearn.preprocessing import StandardScaler

numerical_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Loan_Amount_Term', 'Credit_History']
scaler = StandardScaler()
df[numerical_cols] = scaler.fit_transform(df[numerical_cols])

X = df.drop(['Loan_ID', 'Loan_Status_V'], axis=1)
y = df['Loan_Status_V']

display(X.head())
display(y.head())
```

[illegible]

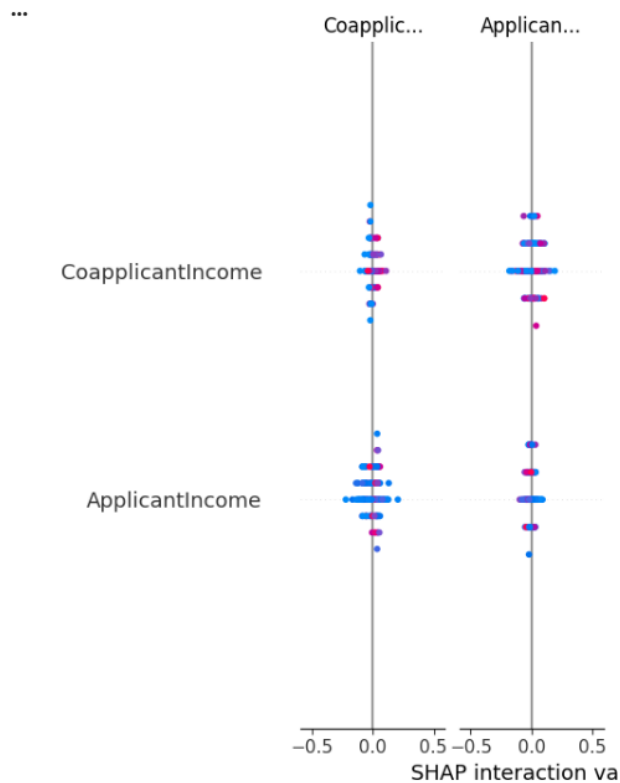
Loan_Status_Y

0	True
1	False
2	True
3	True
4	True

dtype: bool

```
# Calculate SHAP values for the test set
shap_values = explainer.shap_values(X_test)

# Generate a summary plot
shap.summary_plot(shap_values, X_test)
```



What risks arise if AI decisions are a “black box”?

ANS:

- **Lack of trust:** People don't trust what they don't understand.
- **Bias:** Models can make unfair or discriminatory decisions.
- **Hard to fix:** It's difficult to find and correct errors.
- **Legal problems:** Regulations often require knowing how AI decides.
- **Slow improvement:** It's tough to make the model better if you don't know why it works.
- **Ethical issues:** It's hard to be responsible for decisions you can't explain.

How explainability ensures fairness in financial decisions

ANS:

Revealing Bias: It helps identify if decisions are based on protected characteristics (like race, gender, etc.) instead of relevant financial factors.

Enabling Auditing: It allows experts to review the decision-making process for potential discrimination or unfair practices.

Providing Recourse: When a decision is unfavorable, an explanation helps individuals understand why and potentially challenge the outcome if it seems

	unfair. Building Trust: Transparent decision-making fosters trust with customers and regulators.	
	Task 6 – Environmental and Societal Impact Students explore how large-scale AI coding assistance affects energy consumption (cloud servers, GPUs). Discuss: Is there an ethical responsibility to consider sustainability? How should developers balance performance vs environmental cost?	
	Expected Outcomes: • Students evaluate inclusivity, transparency, and accountability in AI coding. • Students identify risks of dual-use and misuse of AI-generated code. • Students understand broader societal and environmental implications of AI-assisted coding. OUTPUT: Is there an ethical responsibility to consider sustainability? • Yes, absolutely. As AI models become larger and more powerful, the energy required to train and run them increases significantly. This has a direct impact on carbon emissions and environmental sustainability. • Developers and organizations have a responsibility to be mindful of this impact and explore ways to mitigate it. This could involve using more energy-efficient hardware, optimizing algorithms, or choosing cloud providers that use renewable energy sources. • Transparency about the energy consumption of AI models is also an ethical consideration, allowing users and the public to understand the environmental cost. How should developers balance performance vs environmental cost? • This is a key challenge. Often, higher performance in AI models comes with increased computational cost and energy consumption. • Strategies for balancing: ◦ Model optimization: Exploring smaller, more efficient models that can achieve similar results with less energy. ◦ Hardware selection: Utilizing specialized hardware designed for energy efficiency. ◦ Responsible deployment: Avoiding unnecessary or redundant AI computations. ◦ Life cycle assessment: Considering the environmental impact throughout the entire AI development and deployment lifecycle, not just during training. ◦ Prioritizing tasks: Determining if the performance gains are truly necessary for the specific application and if the environmental cost is justified.  Deliverables (For All Tasks) 1. AI-generated prompts for code and test case generation. 2. At least 3 assert test cases for each task. 3. AI-generated initial code and execution screenshots. 4. Analysis of whether code passes all tests. 5. Improved final version with inline comments and explanation. 6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.	

